When Big Firms Stumble, Startups Pay: Category Spillover and Startup Responses

Yunhee Han, **Sang-Joon Kim***

Yunhee Han: Graduate Student, School of Business, Ewha Womans University (yunheehan@ewha.ac.kr)
Sang-Joon Kim (Corresponding author): Professor, School of Business, Ewha Womans University (s.kim@ewha.ac.kr)

Abstract

This study examines how a privacy scandal at a large firm affects the fundraising of startups that share its category but had no part in the scandal, and which of these startups bear more or less of the resulting penalty. Drawing on research on legitimacy spillover and organizational stigma, we analyze 51,738 U.S. startups listed in Crunchbase that were founded before (2015–2017) or after (2019–2021) the 2018 Facebook–Cambridge Analytica scandal. Social platform startups were less likely to raise funding than other startups in the same industry groups. The penalty was larger for personal record services, in which users log sensitive information such as health conditions or emotions for their own use, and it disappeared for content sharing services, in which users post content intended to be seen by others. Because content sharing services overlap more with Facebook's core business, this pattern suggests that the size of a spillover depends less on how similar a firm is to the offending firm than on how much harm the same problem would cause if it occurred at that firm. In addition, social platform startups that stated in their descriptions that users can control their own exposure were more likely to raise funding. The study shows that a category is both a source of legitimacy for startups and a channel through which other firms' misconduct spreads, and that startups lacking the risk-management means of established firms can still distinguish themselves within a stigmatized category through service design and self-presentation.

Keywords: legitimacy spillover, stigma transfer, categories, venture funding, privacy, social platform startups

* Corresponding author: Sang-Joon Kim, Professor, School of Business, Ewha Womans University (s.kim@ewha.ac.kr)

I. Introduction

Large-scale breaches of personal data have become a recurring feature of corporate life in the United States. The 2017 Equifax breach exposed the personal information of about 147 million people and ended in a 2019 settlement of up to $700 million with the Federal Trade Commission (FTC), the Consumer Financial Protection Bureau, and the states (Federal Trade Commission, 2019a). That same month, the FTC imposed a $5 billion penalty on Facebook for privacy violations surrounding the Cambridge Analytica scandal (Federal Trade Commission, 2019b). More recently, a 2024 cyberattack on Change Healthcare compromised data on about 190 million people (Infosecurity Magazine, 2025), and U.S. organizations reported 3,158 data compromises in 2024 alone, resulting in more than 1.3 billion victim notices (Identity Theft Resource Center, 2025). When such incidents occur, public attention converges on the large firm responsible, and the subsequent debate centers on how that firm should be punished and regulated.

Yet the consequences of an incident are not confined to the firm that caused it. Evaluators, the audiences such as investors, customers, and the media that judge firms and decide whether to support them, perceive firms through categories, groups of firms regarded as the same kind (Zuckerman, 1999; Vergne & Wry, 2014). As a result, what happens to one member of a category spills over to the others (Kostova & Zaheer, 1999). Spillover can work in members' favor, letting them share in the legitimacy their category has earned, that is, the perception that their actions are appropriate (Suchman, 1995). It can also work against them by spreading stigma, the perception that a firm has a fundamental, deep-seated flaw (Devers, Dewett, Mishina, & Belsito, 2009). When one member commits wrongdoing, blameless firms that merely share its category come under suspicion, a phenomenon known as guilt by association (Jonsson, Greve, & Fujiwara-Greve, 2009). Startups, which lack a track record, are particularly exposed, because evaluators judge them by their category rather than by the firm itself. The category that makes a startup understandable to investors is thus also what exposes it to other firms' misconduct.

Research on this spillover leaves two questions open. The first is what makes a blameless firm close to the offender. Prior research holds that the closer a firm is to the offender, the more of the spillover it bears, and has defined closeness as resemblance to the offender in business or organizational form (Kostova & Zaheer, 1999; Jonsson et al., 2009) or as holding the attribute implicated in the incident (Diestre & Rajagopalan, 2014; Naumovska & Zajac, 2022). Yet firms that hold the same attribute can differ in its depth, the degree to which the attribute, in the form a firm holds it, touches what the affected parties most want protected. Resemblance and depth need not go together, and if evaluators follow depth, the firms that resemble the offender most may not be the ones that suffer most. The second is which firms in a category bear less of a spillover. Spillover arises because evaluators cannot observe each firm and so cannot tell blameless firms from flawed ones (Barnett & King, 2008). It should therefore weaken for firms with high distinguishability, the degree to which evaluators can see from observable information that a firm lacks the flaw that caused the

incident (Barnett & King, 2008). The known sources of such protection, however, are means available to established firms, such as industry self-regulation, endorsements from industry associations, and exit from the stigmatized business (Barnett & King, 2008; Diestre & Rajagopalan, 2014; Piazza & Perretti, 2015), and even an industry-wide self-regulatory program did not protect its members from other firms' accidents more than nonmembers (Barnett & King, 2008). Both questions matter most for startups, yet spillover research has examined them almost entirely among established firms, through effects on their stock prices or customer retention (Barnett & King, 2008; Jonsson et al., 2009; Naumovska & Zajac, 2022). For an established firm, a spillover lowers a market value that it can absorb. For a startup, the same spillover can cut off the funding it needs to survive, and fewer new ventures enter a market that investors avoid (Stinchcombe, 1965; Pontikes & Barnett, 2017). Startups also cannot use the means available to established firms, and because their entire business lies within the category, they cannot exit it. What makes some startups distinguishable to investors, and whether that spares them part of the penalty, remains unknown.

Building on research on categories and the spillovers that occur within them, this study addresses these gaps by examining how an incident at a large firm affects the fundraising of startups in the same category, and which startups within that category bear more or less of the penalty. Our empirical setting is the 2018 Facebook–Cambridge Analytica scandal and social platform startups. We define social platforms as online services in which users post information about themselves and form relationships or interact with other users through that information (boyd & Ellison, 2007; Kane, Alavi, Labianca, & Borgatti, 2014). The scandal was caused by a single large platform, but it made visible the risks of the social platform category as a whole, while most social platform startups at the time had no direct connection to it. Whereas prior research on this scandal examined parties tied to Facebook itself (Zheng & Wang, 2025), we examine startups in the same category that had no part in it and observe the consequences through whether they raised funding, following evidence that category-level changes are reflected in venture investment (Jia, Jin, & Wagman, 2021). The sample consists of 51,738 U.S. startups founded between 2015 and 2021, excluding those founded in 2018, the year of the scandal.

Comparing startups founded in the three years before (2015–2017) and after (2019–2021) the scandal, we find that social platform startups were less likely to raise funding than other startups in the same industry groups (the broad sectors, such as Software, into which Crunchbase groups its industries). This penalty, however, differed by service type. Personal record services, in which users log sensitive information such as health conditions or emotions for their own use, were about 15 percentage points less likely to raise funding than other startups in the same industry classification. In contrast, content sharing services, in which users post videos, photos, and other content intended to be seen by others, were statistically indistinguishable from other startups. Because content sharing services overlap more with Facebook's business, this result runs counter to the prediction of prior research that firms more similar to the offender suffer more. In addition, social platform startups that

stated that users can decide the extent to which they are exposed were more likely to raise funding, even within the same category.

Our central argument is that a category is both a foundation of legitimacy for startups and a channel through which other firms' misconduct spreads. Startups must belong to a category to be understood by investors, yet that very membership exposes them to suspicion even when they are unrelated to the incident. The study makes three contributions. First, it extends the focus of category spillover research from the losses of established firms to the resource acquisition of new ventures. Second, it shows that among firms that hold the implicated attribute, a spillover falls more heavily on those that hold it at greater depth, even when they resemble the offender less, thereby refining the criterion that determines the size of a spillover. Third, it shows that distinguishability, so far examined through the collective means of established firms, also protects startups, and that it can work through what investors read in a startup's self-presentation.

II. Theoretical Background and Hypotheses

2.1 Categories and Legitimacy Spillover

Legitimacy is a generalized perception that an organization's actions are desirable, proper, or appropriate within some socially constructed system of norms, values, and beliefs (Suchman, 1995). Legitimacy is a condition for obtaining external resources, and it matters especially for new ventures that lack a track record and reputation (Aldrich & Fiol, 1994; Zimmerman & Zeitz, 2002). Because evaluators do not yet know what a new venture even does, the venture must gain legitimacy by explaining itself within frames that evaluators can understand (Lounsbury & Glynn, 2001; Navis & Glynn, 2011).

Categories are a prime example of such frames. A category is a group of organizations that evaluators regard as the same kind (Zuckerman, 1999; Vergne & Wry, 2014). Because evaluators cannot understand every organization individually, they understand and compare organizations through categories that group similar ones. Zuckerman (1999) showed that firms that fit poorly into the categories covered by securities analysts receive discounted valuations in the stock market. Investors likewise evaluate startups through categories. Pontikes (2012) showed that venture capitalists' evaluations of software firms depend on how the firms classify themselves, and Wry, Lounsbury, and Jennings (2014) showed that the category from which a nanotechnology startup originates becomes the reference point investors use to understand it. Evaluations of a firm thus depend not only on its own characteristics but also on the category to which it belongs, and the legitimacy of a category becomes a condition its members inherit (Navis & Glynn, 2010).

Kostova and Zaheer (1999) called this legitimacy spillover: because evaluators perceive similar organizations as a single category, a change in the legitimacy of one carries over to the others. This spillover matters most for startups. Lacking a track record, they cannot yet be judged on their own, so investors rely more heavily on judgments about the category when evaluating startups (Aldrich & Fiol,

1994; Zimmerman & Zeitz, 2002), and category membership lets them draw on legitimacy they have not yet earned themselves (Navis & Glynn, 2010).

2.2 Stigma Spillover and Startups

What spills over within a category, however, is not always favorable. If firms inherit the legitimacy of their category, they also inherit its losses when that legitimacy is shaken. When a blameless organization is evaluated negatively because of its association with an offender, it suffers guilt by association (Jonsson et al., 2009), and such negative spillovers tend to be stronger than positive ones (Haack et al., 2014). When a firm causes an incident, evaluators judge similar organizations together and even curtail transactions with blameless firms (Jonsson et al., 2009; Yu, Sengul, & Lester, 2008). Jonsson et al. (2009) showed that after misconduct at the Swedish insurer Skandia was reported, fund companies affiliated with other, blameless insurers also lost customers. Barnett and King (2008) showed that in the U.S. chemical industry, one firm's accident lowered the stock prices of other chemical firms, and that this effect grew after the Bhopal disaster. Zavyalova, Pfarrer, Reger, and Shapiro (2012) showed that product recalls by industry peers make media coverage of a focal firm more negative.

Research on organizational stigma describes the same process from the side of the negative evaluation. Organizational stigma is stakeholders' perception that an organization possesses a fundamental, deep-seated flaw (Devers et al., 2009), and stigma transfer is the process by which that perception moves to connected organizations. The idea that association with a stigmatized party is enough to stigmatize others originates in Goffman's (1963) notion of courtesy stigma and was later extended to the organizational level (Devers et al., 2009). Stigma transfer is thus the negative side of category spillover, and it follows the same logic. Because evaluators generalize one firm's wrongdoing into inferences about the group to which it belongs, the closer a blameless firm appears to the offender, the more of the stigma it receives (Kostova & Zaheer, 1999; Diestre & Rajagopalan, 2014). When incidents recur or receive heavy coverage, negative evaluations harden into a stigma attached to the category itself (Devers et al., 2009; Vergne, 2012), and firms in a stigmatized category come under suspicion even when they are unrelated to the incident (Piazza & Perretti, 2015).

Such spillovers are likely to weigh especially heavily on startups. An incident in the category confronts established firms and startups alike, but the means for managing the resulting risk differ. Established firms can draw on crisis response teams, legal and public relations capabilities, and existing stakeholder relationships to manage the aftermath and shape evaluators' interpretations (Coombs, 2007; Zavyalova et al., 2012). Startups, by contrast, have yet to build such means, and, as the liability of newness suggests (Stinchcombe, 1965), they also lack the resources to compensate. Category-level changes are indeed reflected in investment. Investment in technology ventures declined after the European Union's General Data Protection Regulation (GDPR) took effect (Jia et al., 2021). Visible failures in a market reduce both entry into that market and venture funding (Pontikes &

Barnett, 2017), and venture capitalists that had invested in fraudulent startups reduced new investments in the affected industry after the fraud came to light (Lyu & Shen, 2023). In crowdfunding, too, the successes and failures of earlier projects in a category spilled over to the fundraising of later ones (Soublière & Gehman, 2020; Li, Dong, & Makino, 2023). The spillovers in these studies, however, mostly occurred among ventures of similar size.

An incident caused by a large firm amplifies these effects. A firm that represents its category serves as a reference point that shows evaluators what the category is, so when that firm falters, the legitimacy of the entire category is called into question (Smith, Bergman, & Kreiner, 2021). An incident at a large firm also draws broad attention from the media and regulators, and that coverage reinforces stereotypes about the category as a whole (Naumovska & Zajac, 2022). Investors are then inclined to attribute the same risk even to startups unrelated to the incident. Startups that share a category with the large firm responsible for the incident should therefore be disadvantaged in fundraising by virtue of their category membership alone.

> **Hypothesis 1 (H1).** *Startups in the same category as a large firm that caused a user-information misuse incident, that is, social platform startups, will be less likely to raise funding than other startups.*

2.3 Depth of the Implicated Attribute

Spillovers do not affect all firms in a category equally (Paruchuri & Misangyi, 2015). According to Naumovska and Zajac (2022), when evaluators observe one firm's wrongdoing, they both suspect the entire category to which the firm belongs (inductive generalization) and identify attributes that appear to have caused the wrongdoing, suspecting more strongly the firms that share them (deductive generalization). We refer to such an attribute as the implicated attribute. Even within the same category, therefore, the size of the spillover depends on how closely a firm appears to be connected to the problem the incident revealed. In Jonsson et al. (2009), the loss of legitimacy was greatest among insurers holding real estate, the asset implicated in the misconduct, and Diestre and Rajagopalan (2014) showed that after chemical accidents, investors penalized firms using the same chemical in proportion to their usage. Durand and Vergne (2015) likewise showed that in the arms industry, media attacks on peers had larger effects on firms in the same subcategory.

What, then, makes a firm close to the problem? Prior research has answered in two ways: by overlap with the offender in organizational form, resources, or product markets (Naumovska & Lavie, 2021), or, as in the studies above, by whether and how much a firm holds the attribute implicated in the incident. Overlap with the offender does not necessarily bring a firm closer to the problem; firms whose product markets overlap heavily with the offender's may even gain by taking over its customers and investors (Paruchuri, Pollock, & Kumar, 2019; Naumovska & Lavie, 2021). And firms that hold the same attribute can still differ in how deeply it runs through their business. We define the depth of the implicated attribute as the degree to which the attribute, in the form a firm holds it, touches what

the affected parties most want protected. The same ingredient, for example, touches more of what consumers want protected when it goes into infant formula than when it goes into snacks. Unlike the damage an incident eventually causes, depth can be observed before any incident occurs, from what a firm offers and to whom. Because investors cannot observe startups from the inside, they make this judgment from externally visible attributes, such as the industry in which a firm operates and the service it provides (Pontikes, 2012; Wry et al., 2014). Because the startup still belongs to the troubled category, it continues to face suspicion directed at the category (H1), but the degree to which that suspicion is applied can vary. We argue that this degree rises with the depth of the implicated attribute, even when a firm resembles the offender less than others do.

In the privacy scandal studied here, the implicated attribute is user information, and the problem the incident revealed was that information users had provided was used beyond the scope they expected. Privacy research captures this depth through the sensitivity of information, which it defines as the degree to which people would not want others to know the information they provide, and it shows that sensitivity shapes how willing people are to disclose information and how much risk they perceive in doing so (Mothersbaugh, Foxx, Beatty, & Wang, 2012; Milne, Pettinico, Hajjat, & Markos, 2017). Health information in particular is treated as highly sensitive, both by individuals deciding whether to disclose it online (Bansal, Zahedi, & Gefen, 2010) and by privacy laws that give it special protection (Ohm, 2015). Sensitivity also depends on context: whether a use of information violates privacy depends on how far it departs from the expectations under which the information was shared, a view formalized as contextual integrity (Nissenbaum, 2004; Acquisti, Brandimarte, & Loewenstein, 2015).

All social platforms rely on user information, but they differ in how that information is meant to be used. At one end are personal record services, in which users log information they would prefer others not see, such as their health conditions, emotions, and personal concerns, for their own use. Such information is provided on the premise that it stays on the platform, so the implicated attribute runs deep in these services. When the attribute runs deep, the suspicion directed at the category lands on exactly what users most want protected: if the flaw revealed by the incident were present at such a firm, the information involved would be the information users least wanted others to see. Investors therefore have no reason to discount the category's suspicion for these startups and should instead intensify it.

Hypothesis 2 (H2). *The fundraising disadvantage of social platform startups will be greater for those in which user information runs deep.*

At the other end are content sharing services, in which users post videos, music, photos, and other content intended from the outset to be seen by others. Because users upload content and share it with others, these services overlap most with Facebook's business. They also handle user information and so face suspicion toward the category, but because much of the information is already meant to be

public, the implicated attribute runs shallow in these services. When the attribute runs shallow, the suspicion directed at the category finds little to attach to: even if the same flaw were present, the information involved would largely be what users had chosen to show others. Investors should therefore discount the category's suspicion for these startups, even though they resemble the offender more.

Hypothesis 3 (H3). *The fundraising disadvantage of social platform startups will be smaller for those in which user information runs shallow.*

2.4 Distinguishability within a Category

Depth describes how deeply a firm holds the implicated attribute. Whether evaluators can tell that the flaw behind the incident is absent at a particular firm is a separate question. Spillover arises in the first place because evaluators cannot observe each firm's conduct and so cannot discriminate between blameless and flawed firms in a category (Barnett & King, 2008). We define distinguishability as the degree to which evaluators can see, from observable information, that a firm lacks the flaw that caused the incident. According to Naumovska and Zajac (2022: 373), "if an innocent organization falls under a negative stereotype that causally links the innocent firm with corporate misconduct, then that innocent firm will suffer a greater negative spillover effect, irrespective of its similarity to the offending firm." By the same logic, a firm that visibly lacks the cause of the misconduct should suffer less. Empirical studies support this view (Barnett & King, 2008; Diestre & Rajagopalan, 2014; Piazza & Jourdan, 2018; Smith et al., 2021; Khessina, Reis, & Verhaal, 2021). Responsible Care, the self-regulatory program of the U.S. chemical industry, reduced the spillover of member firms' accidents to other firms (Barnett & King, 2008), and spillovers weakened when an industry association vouched for the safety of the chemical involved in an accident (Diestre & Rajagopalan, 2014). After scandals, organizations that resembled the offender but appeared to adhere to stricter standards benefited (Piazza & Jourdan, 2018), and after a firm that had represented its category collapsed, the remaining firms regained legitimacy by emphasizing how they differed from it (Smith et al., 2021). Research also shows that revealing an organization's identity reduces concerns about stigma transfer (Khessina et al., 2021).

In these studies, the means of attracting less stigma have been examined only among firms that had already secured legitimacy, not among startups. These means include establishing industry self-regulation (Barnett & King, 2008), obtaining endorsements from industry associations (Diestre & Rajagopalan, 2014), and exiting the stigmatized business (Durand & Vergne, 2015; Piazza & Perretti, 2015). Even these collective means made the industry as a whole less vulnerable rather than distinguishing individual members, as Responsible Care did not protect its members from other firms' accidents more than nonmembers (Barnett & King, 2008). Startups lack the influence to shape industry standards, and because their entire business lies within the category, they cannot exit. For a startup to be distinguishable, the information investors first encounter must show that the firm lacks

the flaw. Because evaluations of new ventures rely heavily on how they present themselves (Lounsbury & Glynn, 2001; Martens, Jennings, & Jennings, 2007), the service features stated in a firm's description can serve this role. Such features need not have been adopted in response to the incident, and we do not treat them as a strategic choice. They work as a defense through investors' reading: whatever a founder intended in writing the description, features showing that the flaw is absent make the firm distinguishable and shield it from part of the suspicion directed at the category (Devers et al., 2009; Piazza & Jourdan, 2018).

In our context, the flaw behind the incident was that information users had provided was used beyond their control. User exposure control refers to features that let users decide how much, and to whom, they are visible, such as anonymous or pseudonymous participation, spaces visible only to invited members, and user-defined visibility settings. Startups that state such features in their descriptions show that users decide how far their information is visible, making them distinguishable from the flaw behind the incident. Distinguishability differs from the depth discussed in Section 2.3. A personal record service holds highly sensitive information, yet it can still be distinguishable if its description shows that users control who sees that information. Depth concerns how deeply a firm holds the implicated attribute, whereas distinguishability concerns whether investors can see that the flaw is absent at that firm.

> **Hypothesis 4 (H4).** *Among social platform startups, those that state user exposure control features will be more likely to raise funding than those that do not.*

III. Methods

3.1 Data

We used Crunchbase data to analyze the relationship between an incident caused by a large firm and the subsequent fundraising of startups in the same category. The final sample comprises 51,738 U.S. startups founded between 2015 and 2021. Because these firms span a wide range of industries beyond social platforms, social platform startups can be compared with other startups.

We chose Crunchbase for three reasons. First, it is a leading database that jointly provides startups' founding years, industry classifications, company descriptions, and funding histories, and it has been widely used in research on early-stage fundraising (Dalle, den Besten, & Menon, 2017). Second, it covers not only funded firms but also firms that did not raise funding, which reduces the problem of selecting the sample on the outcome. This matters for our study, because a dataset consisting only of venture-backed firms cannot use whether a firm raised funding as the dependent variable. Third, because it provides self-reported industry classifications and descriptions, we can measure within the same data both the category a firm belongs to and the design features it emphasizes.

We used data on 61,618 U.S. startups founded between 2015 and 2021, and we constructed social platform status, service type, and user exposure control from industry classifications and company

descriptions. In building the sample, we identified a limitation in the founding-date information. About 44% of firms record only a founding year, with the date uniformly entered as January 1, so for firms founded in 2018 it is impossible to tell whether they were founded before or after the Cambridge Analytica scandal (March 2018). We therefore excluded 9,880 firms founded in 2018 and used the 51,738 firms founded before (2015–2017) or after (2019–2021) the scandal as the main sample. The full sample including firms founded in 2018 was used in robustness checks.

3.2 Research Context

We chose social platform startups and the 2018 Cambridge Analytica scandal as our empirical context. This context serves as a case for testing a general, category-level theory; our interest does not lie in a particular industry or event. In March 2018, it was reported that information on about 87 million Facebook users had been passed to a political consulting firm. That Cambridge Analytica had used Facebook user data in political campaigns was first reported in December 2015 (Davies, 2015), and the March 2018 reports, which revealed the scale of the matter, drew worldwide attention (Cadwalladr & Graham-Harrison, 2018). We date the event to March 2018. After the 2015 report, neither the firm nor regulators took official action, whereas in March 2018 Facebook suspended the firm from its platform (Facebook, 2018) and the U.S. Federal Trade Commission (FTC) announced an investigation into Facebook's privacy practices (Federal Trade Commission, 2018). The incident thereby widened from one firm's deviance into a problem of how social platforms in general manage user information. The scandal has also served as the setting for research on how stakeholders adjust their relationships with firms after a scandal (Zheng & Wang, 2025). Although the party to the incident was a single large firm, Facebook, subsequent criticism and regulatory debate extended to social platforms operating on user information in general. Most social platform startups at the time, in contrast, had no direct connection to the incident. This context therefore clearly separates parties to the incident from non-parties (Barnett & King, 2008; Jonsson et al., 2009), allowing us to observe penalties that arise from category membership alone. Furthermore, because social platforms divide into services in which users record information for themselves and services in which users post content for others to see, we can compare differences by service type within the same category.

3.3 Dependent Variable

The dependent variable is whether a startup raised funding (*Funded*). For early-stage startups, external funding is not only a resource needed for survival and growth but also a signal that a key evaluator, the investor, has recognized the firm's legitimacy. If negative evaluations of a category spill over to startups, the effect should appear first in fundraising, where evaluators' judgments are directly reflected. We coded firms with a funding record in Crunchbase as 1 and those without as 0, based on the total funding amount (in U.S. dollars) recorded in Crunchbase. Because firms without a funding record are coded with an amount of 1 rather than 0 in the data, we regarded firms with total funding above one dollar as funded. In the main sample, 15,883 firms (30.7%) raised funding.

3.4 Independent and Moderating Variables

The independent variable is whether a startup is a social platform startup (*Social Platform*). Following the definition in Section II, a social platform is a service in which users post information about themselves and form relationships or interact with other users through it. Because evaluators perceive firms through their categories (Zuckerman, 1999), category membership is appropriately measured by the industry classifications that firms register themselves. We coded as 1 firms whose Crunchbase industry classifications include at least one of Social Network, Social Media, Private Social Networking, or Dating. In the main sample, 1,820 firms (3.5%) meet this criterion.

The moderating variables are the service types of social platforms. As discussed in Section II, social platforms divide into personal record services, in which users keep sensitive information for themselves, and content sharing services, in which users post content for others to see. We measured both types with Crunchbase industry classifications. A firm's registered industry classification is a self-declared marker of the service it provides, and because our hypothesis tests compare social platforms with other startups in the same classification, the classification distinguishes service types within social platforms rather than industry differences as such. *Personal Record* equals 1 if a firm is classified in at least one of Wellness, Fitness, Mental Health, Health Care, Personal Health, Health Diagnostics, or Medical. Social platforms in this group are services in which users record and manage their health conditions, emotions, and concerns or share them with people in similar situations. *Content Sharing* equals 1 if a firm is classified in Media and Entertainment. Social platforms in this group are services in which users post videos, music, photos, and other content for other users to see. Among social platform startups, 71 are personal record services and 101 are content sharing services, and no firm falls into both.

We used company descriptions to verify that these classifications indeed separate the two service types. The share of descriptions containing expressions of self-recording (e.g., track your, journal, mood, symptom) or sensitive information (e.g., mental health, therapy, support group, patient) was 28.2% among personal record social platforms, far higher than among content sharing services (4.0%) or other social platforms (2.6%). Conversely, the share containing expressions of showing content to others (e.g., video, music, stream, creator, follower) was 67.3% among content sharing services, higher than among personal record services (19.7%) or other social platforms (31.1%) (Appendix Table A6). The two classifications thus match the ways of using data revealed in the descriptions. Because the difference between the two types is also a matter of degree, we additionally report results using description-based measures for individual firms in Section 4.2.

The explanatory variable for H4 is *User Exposure Control*, which captures whether a platform is designed so that users can decide how much, and to whom, they are visible. Because the problem in the incident was that users could not control how far their information spread, we measured designs in which users themselves set the scope of their exposure. We excluded the ways in which platforms verify users or moderate content, because in those cases the platform, not the user, sets the scope of

exposure. Because such features become apparent when users use the service and are what startups display when presenting themselves externally, we measured them using company descriptions and cross-validated the measure against each firm's website (for the use of descriptions to measure founding strategy, see Guzman & Li, 2023). The measurement proceeded in three steps. First, before seeing any results, we defined lists of expressions for three types: anonymous or pseudonymous participation, private communities, and visibility settings. Next, we read the actual descriptions of firms captured by these expressions and removed expressions captured in contexts unrelated to the concept. Finally, we fixed the list and generated the variable only once. Crunchbase industry classifications are self-registered markers, and among them, Private Social Networking directly indicates a product design in which the platform operates privately. Because comparisons are made only among social platforms, this classification distinguishes a design type rather than industry membership, so we also coded firms in this classification as 1. In total, 121 social platform startups (6.6%) are coded as 1. The full list of expressions is reported in Appendix Table A3, and definitions of all variables are reported in Appendix Table A1.

3.5 Control Variables

We controlled for firm characteristics that may affect fundraising. Patents signal early-stage firms' technological capability and quality to investors and are associated with fundraising (Hsu & Ziedonis, 2013); we therefore included the log of the number of patents (*ln # Patents*). Because venture capital is geographically concentrated and investors tend to invest in nearby firms (Chen, Gompers, Kovner, & Lerner, 2010), we included indicators for location in Silicon Valley and New England. Because founders' gender and race may affect investors' evaluations (Brooks, Huang, Kearney, & Murray, 2014), we included indicators for female and white founders. Because a firm's social orientation may be read by investors as a different signal, we included indicators for nonprofit status and participation in Pledge 1%. In addition, because firms that span multiple categories are harder for evaluators to identify and may be evaluated differently (Zuckerman, 1999; Hsu, Hannan, & Koçak, 2009), we controlled for the number of industry classifications a firm registered in Crunchbase (*# Industry Categories*).

We also included the main effects of personal record and content sharing services in all models so that the interaction effects are not confounded with the effects of the service types themselves. Next, we included founding-year fixed effects. These absorb year-to-year differences in the investment market as well as differences in right-censoring, whereby more recently founded firms had less time to raise funding before data collection.

Finally, we included Crunchbase industry-group fixed effects. The unit of analysis remains the individual firm; the fixed effects only determine which firms each firm is compared with. Crunchbase groups its detailed industries into broader industry groups, and a firm may belong to several groups at once. Social platform startups span multiple groups such as software, mobile, and content, and without

fixed effects they would also be compared with firms in groups such as biotechnology and hardware, whose funding rates differ structurally. We therefore included a dummy for each industry group a firm belongs to, so that, for example, a social platform startup in the Software and Mobile groups is compared with other startups in those same groups. H1 thus tests not whether social platforms receive less funding than other industries, but whether, within the same industry groups, firms in the social platform category receive less funding than firms outside it. We excluded the four industry groups that overlap with the definitions of social platforms and the two service types (Internet Services, Media and Entertainment, Health Care, and Community and Lifestyle), because including them would absorb the very effects under study (43 industry groups in total). Results of models without industry-group fixed effects are reported in Appendix Table A5.

3.6 Analysis

Because the dependent variable is binary, we used logistic regression. All models include the control variables, industry-group fixed effects, and founding-year fixed effects. We first estimated a model with only the control variables (Model 1) and then added social platform status to test H1 (Model 2). Next, we added the interaction between social platform status and each of the two service types, one at a time, to test H2 and H3 (Models 3–4). We entered the interactions separately in order to examine and compare how each type changes the category penalty. H4 was estimated only among social platform startups (Models 5–7). Because this analysis asks which startups within the same category bear less of the penalty, we compared firms within that category.

In logit models, the size and significance of an interaction coefficient alone do not allow a straightforward interpretation of the effect (Ai & Norton, 2003; Hoetker, 2007). We therefore report odds ratios and predicted probabilities of raising funding by group alongside the coefficients. Finally, we repeated the analysis on the full sample of 61,618 firms, including those founded in 2018.

IV. Results

Table 1 reports descriptive statistics and correlations for the main variables. In the main sample, 30.7% of firms raised funding, and 1,820 firms (3.5%) were social platform startups. Among social platform startups, 71 were personal record services and 101 were content sharing services, and 121 (6.6%) stated user exposure control features. The correlations with fundraising were largest for the number of patents (r = 0.18) and the number of industry categories (r = 0.15). Correlations among the variables were generally low. The mean VIF was 1.10, indicating little concern about multicollinearity.

Table 1. Descriptive Statistics and Correlations

Variable	Mean	SD	Min	Max	1	2	3	4	5	6	7	8	9	10	11
1. Funded (0/1)	0.307	0.461	0	1											
2. Social platform (0/1)	0.035	0.184	0	1	−0.02										
3. Personal record (Wellness/Health)	0.131	0.337	0	1	0.10	−0.05									
4. Content sharing (Media & Ent.)	0.017	0.127	0	1	0.01	0.06	−0.05								
5. Nonprofit (0/1)	0.010	0.102	0	1	−0.04	−0.00	−0.01	−0.00							
6. Pledge 1%	0.003	0.056	0	1	0.02	−0.00	−0.01	−0.00	−0.01						
7. # Industry categories	3.187	1.887	0	23	0.15	0.12	0.04	0.07	−0.04	0.04					
8. ln # patents	0.082	0.406	0	10.5	0.18	−0.02	0.09	−0.01	−0.02	−0.00	0.05				
9. Female founder	0.110	0.313	0	1	0.11	0.03	0.07	0.01	0.01	0.04	0.10	0.02			
10. White founder	0.978	0.145	0	1	−0.09	−0.02	−0.01	−0.02	0.01	−0.01	−0.07	0.00	−0.13		
11. Silicon Valley	0.065	0.247	0	1	0.04	0.00	−0.02	−0.01	−0.01	−0.01	0.05	0.07	0.01	0.00	
12. New England	0.059	0.236	0	1	0.05	−0.02	0.05	−0.01	0.00	0.02	−0.01	0.05	0.01	0.02	−0.07

Note: N = 51,738 (main sample). The mean VIF is 1.10.

4.1 Main Results

Table 2 reports the tests of H1, H2, and H3. H1 predicted that social platform startups would be less likely to raise funding than other startups. Model 2 supports this prediction. Even after controlling for industry-group fixed effects, the coefficient of social platform status was negative and significant ($\beta = -0.248$, $p < 0.001$). All else equal, the odds that a social platform startup raised funding were about 22% lower than those of other startups in the same industry groups, and the average marginal effect indicates a 4.5 percentage-point lower probability of raising funding. This corresponds to about 15% of the sample's mean funding rate (30.7%).

H2 predicted that the disadvantage of social platform startups would be greater for personal record services, in which users keep sensitive information for themselves. Model 3 supports this prediction. The interaction between social platform status and personal record services was negative and significant ($\beta = -0.640$, $p < 0.05$). In terms of predicted probabilities, other startups in the personal record classification had a 35.1% probability of raising funding, whereas social platform startups in the same classification had 20.1%, a gap of 15.0 percentage points (95% confidence interval: −22.8 to −7.1 percentage points). Among other social platform startups, the corresponding gap was 3.9 percentage points. Because the classification itself was favorable for fundraising, this penalty can be interpreted as arising not from the field but from the combination of suspicion toward the category and a service that relies on sensitive information users entrust for their own use.

H3 predicted that the disadvantage of social platform startups would be smaller for content sharing services, in which users post content for others to see. Model 4 supports this prediction. The interaction was positive and significant ($\beta = 0.544$, $p < 0.05$), and its magnitude more than offset the main effect of social platform status (−0.282). The predicted funding probability of content sharing social platform startups was 37.4%, not lower than that of other startups in the same classification (32.1%; difference 5.2 percentage points, 95% confidence interval: −3.9 to 14.4 percentage points).

Given that other social platform startups were 5.1 percentage points less likely to raise funding, the penalty for services premised on sharing shrank enough to become statistically indistinguishable from other startups. Taken together, within the social platform category, the penalty was greatest for personal record services and disappeared for content sharing services.

Table 2. Results of Hypothesis Tests (H1–H3)

Variable	Model 1	Model 2	Model 3	Model 4
Nonprofit (0/1)	−1.1555***	−1.1562***	−1.1538***	−1.1568***
	(0.1425)	(0.1425)	(0.1425)	(0.1425)
Pledge 1%	0.5368**	0.5319**	0.5310**	0.5322**
	(0.1693)	(0.1693)	(0.1693)	(0.1693)
# Industry categories	0.1823***	0.1888***	0.1889***	0.1888***
	(0.0086)	(0.0087)	(0.0087)	(0.0087)
ln # patents	0.8017***	0.8001***	0.8002***	0.8000***
	(0.0308)	(0.0308)	(0.0308)	(0.0308)
Female founder	0.5658***	0.5679***	0.5687***	0.5686***
	(0.0311)	(0.0311)	(0.0311)	(0.0311)
White founder	−0.8992***	−0.9021***	−0.9033***	−0.9046***
	(0.0655)	(0.0656)	(0.0656)	(0.0656)
Silicon Valley	0.1714***	0.1717***	0.1717***	0.1715***
	(0.0401)	(0.0401)	(0.0401)	(0.0401)
New England	0.2457***	0.2430***	0.2437***	0.2429***
	(0.0419)	(0.0419)	(0.0419)	(0.0419)
Personal record (Wellness/Health)	0.2580***	0.2429***	0.2499***	0.2418***
	(0.0339)	(0.0341)	(0.0343)	(0.0341)
Content sharing (Media & Ent.)	0.1309	0.1307	0.1284	0.0669
	(0.0802)	(0.0803)	(0.0803)	(0.0855)
Social platform (0/1)		−0.2481***	−0.2180***	−0.2818***
		(0.0596)	(0.0607)	(0.0617)
Social platform × Personal record			−0.6402*	
			(0.2841)	
Social platform × Content sharing				0.5436*
				(0.2352)
Constant	−0.7597***	−0.7502***	−0.7513***	−0.7460***
	(0.0714)	(0.0714)	(0.0714)	(0.0715)
Industry-group FE	Yes	Yes	Yes	Yes
Founding-year FE	Yes	Yes	Yes	Yes
Observations	51,738	51,738	51,738	51,738
Log likelihood	−28,960	−28,951	−28,948	−28,948

Note: Logistic regression. Standard errors in parentheses. All models include 43 industry-group fixed effects (excluding the four groups that overlap with the variable definitions) and founding-year fixed effects. N = 51,738. $^{+}p < 0.10$, $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

Table 3 reports the test of H4 among social platform startups only. H4 predicted that social platform startups stating user exposure control features would be more likely to raise funding. In Model 6, which includes industry-group and founding-year fixed effects and the control variables, the coefficient of user exposure control was positive and significant ($\beta = 0.438$, $p = 0.042$). The odds that startups stating these features raised funding were about 55% higher than those of other startups. In

terms of predicted probabilities, social platform startups with this design had a 33.4% probability of raising funding, compared with 25.2% for those without, a difference of about 8.2 percentage points. In Model 7, which controls for description length, the coefficient remained in the same direction ($\beta = 0.367$, $p < 0.10$).

Table 3. Results of the H4 Test (Social Platform Startups)

Variable	Model 5	Model 6	Model 7
Nonprofit (0/1)	0.1178	0.0955	0.0708
	(0.6097)	(0.6131)	(0.6134)
Pledge 1%	−0.6859	−0.6291	−0.7198
	(1.1798)	(1.1797)	(1.1744)
# Industry categories	0.0931*	0.0874*	0.0797+
	(0.0414)	(0.0416)	(0.0417)
ln # patents	0.6981**	0.7197**	0.7351**
	(0.2455)	(0.2472)	(0.2479)
Female founder	0.3090*	0.2972+	0.2914+
	(0.1530)	(0.1533)	(0.1535)
White founder	−1.2099***	−1.2199***	−1.2183***
	(0.2807)	(0.2814)	(0.2815)
Silicon Valley	0.0847	0.0706	0.0707
	(0.2227)	(0.2235)	(0.2235)
New England	−0.3535	−0.3661	−0.3292
	(0.3521)	(0.3521)	(0.3526)
Personal record (Wellness/Health)	−0.2529	−0.2729	−0.3082
	(0.3132)	(0.3131)	(0.3142)
Content sharing (Media & Ent.)	0.6825**	0.6963**	0.7031**
	(0.2329)	(0.2333)	(0.2336)
User exposure control (0/1)		0.4378*	0.3671+
		(0.2156)	(0.2189)
Description length (1,000 chars)			0.2578*
			(0.1244)
Constant	−0.2819	−0.2973	−0.3846
	(0.3200)	(0.3206)	(0.3236)
Industry-group FE	Yes	Yes	Yes
Founding-year FE	Yes	Yes	Yes
Observations	1,804	1,804	1,804
Log likelihood	−961	−959	−957

Note: Logistic regression. Standard errors in parentheses. User exposure control follows the main definition (121 firms). Firms in founding-year or industry-group dummies that perfectly predict the outcome are dropped from estimation. N = 1,804. $^{+}p < 0.10$, $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

4.2 Robustness Checks

Results before adding industry-group fixed effects are reported in Appendix Table A5. Without industry-group controls, the social platform penalty appears larger, at 7.5 percentage points (linear probability model), and the results for H2 and H3 are the same in both specifications. In the linear probability model with industry-group fixed effects, social platform startups were 4.9 percentage

points less likely to raise funding ($p < 0.001$), and the interactions with personal record services ($\beta = -0.127$, $p < 0.05$) and content sharing services ($\beta = 0.110$, $p < 0.05$) were also significant.

In the full sample including firms founded in 2018, the results for H1 and H2 held, and H3 and H4 were in the same direction (Appendix Table A2). Excluding Dating from the definition of social platforms, or including only Social Network and Social Media, the results for H1, H2, and H3 held and H4 was in the same direction (Appendix Table A4).

Measuring service type with expressions in individual firms' descriptions instead of industry classifications yielded results in the same direction (Appendix Table A6). The penalty was significantly smaller for social platforms whose descriptions contained expressions of showing content to others (interaction $\beta = 0.249$, $p < 0.05$). The interaction for social platforms whose descriptions contained self-recording or sensitive-information expressions was negative, as predicted ($\beta = -0.235$, n.s.).

Looking at funding amount instead of whether a firm raised funding also yielded results in the same direction. Among the 15,883 funded firms, with the log of total funding as the dependent variable, social platform startups raised about 57% less than other startups in the same industry groups ($\beta = -0.856$, $p < 0.001$; linear regression with robust standard errors, including the control variables, industry-group fixed effects, and founding-year fixed effects).

We also checked the H4 result in several ways. In the linear probability model, the coefficient of user exposure control was positive (0.083, $p < 0.10$), and it was in the same direction under a broader definition that also includes user verification and content moderation expressions (124 firms). The result held when the private community or visibility setting expressions were dropped one at a time.

4.3 Additional Analyses

4.3.1 Are Exposure Control Features a Response to the Incident?

If exposure control features had been added in response to the incident, their prevalence should have risen among firms founded after it. The adoption rate (the share of social platform startups stating user exposure control features) was 6.7% among firms founded before the incident (2015–2017) and 6.4% among those founded after it (2019–2021), showing no increase, and the size of the effect did not differ significantly between the two groups (interaction $\beta = -0.249$, $p = 0.660$). The features thus did not become more common after the incident.

4.3.2 Alternative Explanations: Kill Zones and Privacy Regulation

The penalty for social platform startups might stem from market structure rather than the scandal. Kamepalli, Rajan, and Zingales (2020) showed that venture investment in startups in adjacent fields declines after large platforms such as Facebook or Google acquire firms in those fields, a phenomenon they called the kill zone. If investors avoid markets in which they would compete with large platforms, social platform startups would receive less funding regardless of suspicion toward the category.

Our results, however, are difficult to explain by this account alone. First, under the kill-zone account, the penalty should vary with the likelihood of competing with large platforms and should be unrelated to the sensitivity of the information required from users. Yet the penalty was greatest for personal record services, to which users entrust sensitive information (H2), and content sharing services, which overlap more with Facebook's core business, were indistinguishable from other startups (H3). Second, within the same category, firms stating that users can set their own exposure were more likely to raise funding (H4). Such features are unlikely to reduce the risk of competing with large platforms, but they are consistent with the stigma transfer account in that they show the firm does not have the flaw behind the incident. The influence of market structure cannot be ruled out entirely, but the pattern of results fits better with our explanation that suspicion spreads differently depending on the harm the same problem would cause at a firm and on the likelihood that it occurs there.

Privacy regulation in the same period could also serve as an alternative explanation. The GDPR took effect in May 2018, two months after the scandal, and was followed by declines in investment in data-reliant technology ventures and in app market entry (Jia et al., 2021; Janßen, Kesler, Kummer, & Waldfogel, 2022). These effects, however, appeared mainly among European ventures, and Jia et al. (2021) used U.S. ventures as the comparison group. Our sample consists of U.S. startups, and because comparisons are made between social platforms and other startups within the same industry groups, regulatory costs common to an industry group are largely absorbed by the fixed effects. Moreover, because regulatory costs fall on all firms handling user data, the regulatory account alone has difficulty explaining why the penalty shrank markedly for content sharing services within the same social platform category (H3).

V. Discussion and Conclusion

5.1 Discussion

Prior research has shown that one firm's incident spreads to other firms in the same category, but it has located the damage mainly in the stock prices and customer retention of established firms (Barnett & King, 2008; Jonsson et al., 2009; Paruchuri & Misangyi, 2015). As a result, little has been known about how a large firm's incident affects startups in the same category and which of them bear less of the penalty. We argued that a large firm's incident spreads through the category even to the fundraising of startups unrelated to the incident, that its size depends on the harm the same problem would cause, and that firms showing that they lack the flaw behind the incident bear less of the penalty.

The results support this argument. Social platform startups were less likely to raise funding than other startups in the same industry groups (H1); this penalty was greater for personal record services (H2) and shrank for content sharing services to the point of being statistically indistinguishable from other startups (H3). In addition, within the same category, startups stating user exposure control features were more likely to raise funding (H4). Each result can be interpreted as follows.

First, the H1 result suggests that category membership can be both a foundation of legitimacy and a channel of cost for startups. Social platform startups received less funding even though they were unrelated to the incident and were compared with other startups in the same industry groups. This implies that as long as investors use categories as frames for understanding firms (Zuckerman, 1999; Pontikes, 2012), suspicion attached to those frames also enters their evaluations of firms. Because startups without risk-management means or a track record have no mechanism to filter out this suspicion, spillovers may surface more directly in startups' resource acquisition than in that of established firms.

Second, the H2 and H3 results show the criterion by which evaluators distinguish firms within a category. Both service types were social platforms, yet the penalty moved in opposite directions. This suggests that evaluators' suspicion does not attach uniformly to the category label but is adjusted according to how much harm the risk revealed by the incident would cause if it recurred in a firm's service. In personal record services, a leak of information directly violates users' expectations, so investors likely regarded these services as the businesses most deeply exposed to the same risk as the incident. In content sharing services, by contrast, the spread of information is the purpose of the service, so the same incident was not read as the same risk. This result suggests that the deductive generalization described by Naumovska and Zajac (2022) can operate on the basis of the harm the same problem would cause, not only on organizational form or resources. It also suggests that contextual integrity (Nissenbaum, 2004), which views privacy harm as a mismatch between information and context, can operate not only in consumer reactions but also as a criterion by which investors distinguish firms within a category. Notably, content sharing social platforms were not disadvantaged even though they more closely resembled Facebook's core business. What matters to evaluators, then, is not how closely a firm resembles the offender but how much harm the situation at issue in the incident could cause at that firm.

Third, the H4 result shows to whom stigma transfers less within a category. Startups stating user exposure control features showed that users decide how far their information is visible, which is inconsistent with the flaw behind the incident, namely that users' information was used beyond their control. Investors recognized this difference. The result shows that the logic of stigma transfer research, which holds that evaluators transfer less stigma to organizations that appear free of the flaw (Devers et al., 2009; Hudson & Okhuysen, 2009), and of deductive generalization (Naumovska & Zajac, 2022) also operates in startup investment. Moreover, because these features did not increase after the incident (Section 4.3.1), the result suggests that attracting less stigma does not necessarily require a response tailored to the incident; service features visible to investors can by themselves make a difference. In this respect, H2–H3 and H4 reflect different judgments: H2 and H3 concern the size of the harm should the same problem occur, and H4 concerns the likelihood that the problem occurs at that firm.

Finally, this study did not directly examine which strategies startups adopted after the incident. The results nonetheless suggest that startups in the same category do not simply absorb the penalty from a large firm's incident. The grounds on which investors distinguished firms within the category were how a service handles user information and how the firm presents this to investors, both of which founders can decide. New ventures can gain legitimacy not only by accepting their environment but by choosing how to build and present themselves (Zimmerman & Zeitz, 2002), and the way founders present their ventures affects fundraising (Martens et al., 2007). In this light, even when an entire category is under suspicion, startups can respond actively by showing, through service design and self-presentation, that they do not have the flaw behind the incident. Future research could examine whether startups actually change their service design or self-presentation after such incidents, and whether those changes lead to funding, thereby observing such responses directly.

5.2 Conclusion

This study examined how a large firm's incident affects the fundraising of startups in the same category, and which startups within that category bear more or less of the penalty. The large firm caused the incident, but startups in the same category that were unrelated to it were also disadvantaged in fundraising. That disadvantage, however, was not the same for all startups. Content sharing services, which overlap more with Facebook's business, were indistinguishable from other startups, whereas personal record services, for which the same problem would be more harmful, were the most disadvantaged. In addition, startups stating that users can set their own exposure were evaluated more favorably even within the same category. A category is a starting condition that startups cannot choose, but evaluations within the same category differ according to what investors see in each firm. Startups thus need not passively accept the suspicion that falls on their category; they can respond actively through how they design and communicate their services.

5.2.1 Theoretical Contributions

The central contribution of this study is to show that a category is, for startups, both a foundation of legitimacy and a channel through which other firms' misconduct spreads. Category research has mainly viewed category membership as a means of gaining legitimacy (Navis & Glynn, 2010; Pontikes, 2012), while spillover research has documented the losses arising from that membership mainly among established firms. We show in new ventures that both sides arise simultaneously from the same membership, and this view yields two contributions to category spillover research and one to stigma transfer research. First, we extend the focus of category spillover research from the losses of established firms to the resource acquisition of new ventures. Prior research documented spillovers in the stock prices of listed firms (Barnett & King, 2008; Diestre & Rajagopalan, 2014) and the customer losses of incumbents (Jonsson et al., 2009). Research on new-venture fundraising showed that the successes and failures of earlier crowdfunding projects affect the fundraising of later projects in the same category (Soublière & Gehman, 2020; Li et al., 2023). We show that after an incident at a large

firm representing the category, startups in that category, which have neither a track record nor a reputation, are disadvantaged in fundraising. Because startups depend more heavily on evaluations of their category (Aldrich & Fiol, 1994), the costs of category spillover may concentrate on firms that have only just entered the market. Future research could examine whether this disadvantage carries over to growth and survival after fundraising, and which firms consequently survive within the category.

Second, we revisit the criterion for who suffers more from a spillover. Prior research showed that legitimacy loss spreads more to firms similar to the offender (Jonsson et al., 2009; Diestre & Rajagopalan, 2014), measuring similarity mainly by overlap in organizational form, resources, and product markets (Naumovska & Lavie, 2021). By this criterion, content sharing services, which overlap most with Facebook's business, should have been the most disadvantaged. The results showed the opposite. Content sharing services were indistinguishable from other startups, and personal record services, which overlap less, were the most disadvantaged. This is consistent with the explanation that evaluators distinguish firms within a category not by similarity to the offender but by how much harm the same problem would cause at each firm. We thereby suggest that the criterion of deductive generalization (Naumovska & Zajac, 2022) can shift from similarity to the offender toward relevance to the harm the incident revealed, and we connect contextual integrity (Nissenbaum, 2004) with category spillover research.

Third, we add the case of firms without means of response to stigma transfer research. Stigma transfer research holds that evaluators transfer less stigma to organizations that appear free of the flaw (Devers et al., 2009; Hudson & Okhuysen, 2009), and it has mainly examined means available to established organizations, such as industry self-regulation (Barnett & King, 2008) and exit from the stigmatized business (Durand & Vergne, 2015; Piazza & Perretti, 2015). Startups lack these means and cannot leave their category. We show that such startups still attract less stigma when their self-presentation shows that they lack the flaw behind the incident. Reducing stigma transfer thus depends not only on an organization's resources or influence but also on what appears in the information investors first encounter. Compared with research showing that revealing an organization's identity reduced concerns about stigma transfer (Khessina et al., 2021), what made a difference in our study was not organizational identity but a service feature that lets users be less exposed. This suggests that effective ways of distinguishing oneself depend on the source of the stigma. Future research could examine how the value of such distinctions changes as a category's reputation recovers.

5.2.2 Practical Implications

For founders, this study shows that a category penalty is not fixed. When an entire category comes under suspicion because of a large firm's incident, founders should first consider whether their service is one in which users record information for themselves or one in which users post content for others to see. For services that hold users' sensitive information, building features that let users set

their own exposure and communicating them clearly can be a way to stand apart within the category. In our data, the predicted funding probability of social platform startups stating such features was about 8.2 percentage points higher than that of other social platform startups.

For investors, the study shows that uniformly discounting an entire category can mean missed opportunities. Even among social platforms, services premised on sharing and services in which users set their own exposure differ from the average risk of the category. Investors who look at business structure rather than category membership may find startups that others have passed over. For policy, just as Responsible Care reduced industry-wide damage in the chemical industry (Barnett & King, 2008), certification or disclosure schemes that make design-level safeguards externally verifiable could help limit the damage that spreads to blameless startups.

5.2.3 Limitations and Future Research

Several features of our design point to directions for future research. Our data come from Crunchbase, and the dependent variable captures whether a funding round was recorded; future work that tracks funding amounts, stages, and timing could show at which stage of fundraising a category penalty emerges. We studied one event, one category, and U.S. startups; testing the same question in other categories that have experienced an incident at a leading firm, such as healthcare diagnostics or crypto-asset startups, would establish the scope of the findings. User exposure control was measured from descriptions observed at the time of data collection; archived websites, app store descriptions, or privacy policies could capture product features at founding more directly. Finally, we inferred investors' interpretations from funding outcomes; interviews or experiments with investors would allow this judgment process to be observed directly.

References

Acquisti, A., Brandimarte, L., & Loewenstein, G. 2015. Privacy and human behavior in the age of information. *Science*, *347*(6221): 509-514.

Ai, C., & Norton, E. C. 2003. Interaction terms in logit and probit models. *Economics Letters*, *80*(1): 123-129.

Aldrich, H. E., & Fiol, C. M. 1994. Fools rush in? *The institutional context of industry creation. Academy of Management Review*, *19*(4): 645-670.

Bansal, G., Zahedi, F. M., & Gefen, D. 2010. The impact of personal dispositions on information sensitivity, privacy concern and trust in disclosing health information online. *Decision Support Systems*, *49*(2): 138-150.

Barnett, M. L., & King, A. A. 2008. Good fences make good neighbors: A longitudinal analysis of an industry self-regulatory institution. *Academy of Management Journal*, *51*(6): 1150-1170.

boyd, d. m., & Ellison, N. B. 2007. Social network sites: Definition, history, and scholarship. *Journal of Computer-Mediated Communication*, *13*(1): 210-230.

Brooks, A. W., Huang, L., Kearney, S. W., & Murray, F. E. 2014. Investors prefer entrepreneurial ventures pitched by attractive men. *Proceedings of the National Academy of Sciences*, *111*(12): 4427-4431.

Cadwalladr, C., & Graham-Harrison, E. 2018. Revealed: 50 million Facebook profiles harvested for Cambridge Analytica in major data breach. The Guardian, March 17.

Chen, H., Gompers, P., Kovner, A., & Lerner, J. 2010. Buy local? *The geography of venture capital. Journal of Urban Economics*, *67*(1): 90-102.

Coombs, W. T. 2007. Protecting organization reputations during a crisis: The development and application of situational crisis communication theory. *Corporate Reputation Review*, *10*(3): 163-176.

Dalle, J.-M., den Besten, M., & Menon, C. 2017. Using Crunchbase for economic and managerial research. OECD Science, Technology and Industry Working Papers, No. 2017/08.

Davies, H. 2015. Ted Cruz using firm that harvested data on millions of unwitting Facebook users. The Guardian, December 11.

Devers, C. E., Dewett, T., Mishina, Y., & Belsito, C. A. 2009. A general theory of organizational stigma. *Organization Science*, *20*(1): 154-171.

Diestre, L., & Rajagopalan, N. 2014. Toward an input-based perspective on categorization: Investor reactions to chemical accidents. *Academy of Management Journal*, *57*(4): 1130-1153.

Durand, R., & Vergne, J.-P. 2015. Asset divestment as a response to media attacks in stigmatized industries. *Strategic Management Journal*, *36*(8): 1205-1223.

Facebook. 2018. Suspending Cambridge Analytica and SCL Group from Facebook. Meta Newsroom, March 16.

Federal Trade Commission. 2018. Statement by the Acting Director of FTC's Bureau of Consumer Protection regarding reported concerns about Facebook privacy practices, March 26.

Federal Trade Commission. 2019a. Equifax to pay $575 million as part of settlement with FTC, CFPB, and states related to 2017 data breach. Press release, July 22.

Federal Trade Commission. 2019b. FTC imposes $5 billion penalty and sweeping new privacy restrictions on Facebook. Press release, July 24.

Goffman, E. 1963. Stigma: Notes on the management of spoiled identity. Englewood Cliffs, NJ: Prentice-Hall.

Guzman, J., & Li, A. 2023. Measuring founding strategy. *Management Science*, *69*(1): 101-118.

Haack, P., Pfarrer, M. D., & Scherer, A. G. 2014. Legitimacy-as-feeling: How affect leads to vertical legitimacy spillovers in transnational governance. *Journal of Management Studies*, *51*(4): 635-666.

Hoetker, G. 2007. The use of logit and probit models in strategic management research: Critical issues. *Strategic Management Journal*, *28*(4): 331-343.

Hsu, D. H., & Ziedonis, R. H. 2013. Resources as dual sources of advantage: Implications for valuing entrepreneurial-firm patents. *Strategic Management Journal*, *34*(7): 761-781.

Hsu, G., Hannan, M. T., & Koçak, Ö. 2009. Multiple category memberships in markets: An integrative theory and two empirical tests. *American Sociological Review*, *74*(1): 150-169.

Hudson, B. A., & Okhuysen, G. A. 2009. Not with a ten-foot pole: Core stigma, stigma transfer, and improbable persistence of men's bathhouses. *Organization Science*, *20*(1): 134-153.

Identity Theft Resource Center. 2025. 2024 annual data breach report. January 28.

Infosecurity Magazine. 2025. Change Healthcare breach almost doubles in size to 190 million victims. January 27.

Janßen, R., Kesler, R., Kummer, M. E., & Waldfogel, J. 2022. GDPR and the lost generation of innovative apps. NBER Working Paper No. 30028.

Jia, J., Jin, G. Z., & Wagman, L. 2021. The short-run effects of the General Data Protection Regulation on technology venture investment. *Marketing Science*, *40*(4): 661-684.

Jonsson, S., Greve, H. R., & Fujiwara-Greve, T. 2009. Undeserved loss: The spread of legitimacy loss to innocent organizations in response to reported corporate deviance. *Administrative Science Quarterly*, *54*(2): 195-228.

Kamepalli, S. K., Rajan, R., & Zingales, L. 2020. Kill zone. NBER Working Paper No. 27146.

Kane, G. C., Alavi, M., Labianca, G., & Borgatti, S. P. 2014. What's different about social media networks? *A framework and research agenda. MIS Quarterly*, *38*(1): 275-304.

Khessina, O. M., Reis, S., & Verhaal, J. C. 2021. Stepping out of the shadows: Identity exposure as a remedy for stigma transfer concerns in the medical marijuana market. *Administrative Science Quarterly*, *66*(3): 569-611.

Kostova, T., & Zaheer, S. 1999. Organizational legitimacy under conditions of complexity: The case of the multinational enterprise. *Academy of Management Review*, *24*(1): 64-81.

Li, M. Y., Dong, C. C., & Makino, S. 2023. Does a past category's success influence existing entrepreneurial fundraising? *A legitimacy spillover perspective. Entrepreneurship Theory and Practice*, *47*(6): 2263-2292.

Lounsbury, M., & Glynn, M. A. 2001. Cultural entrepreneurship: Stories, legitimacy, and the acquisition of resources. *Strategic Management Journal*, *22*(6–7): 545-564.

Lyu, Y., & Shen, L. 2023. Once bitten, twice shy: Evidence from venture capital and scam startups. ECGI Working Paper.

Martens, M. L., Jennings, J. E., & Jennings, P. D. 2007. Do the stories they tell get them the money they need? *The role of entrepreneurial narratives in resource acquisition. Academy of Management Journal*, *50*(5): 1107-1132.

Milne, G. R., Pettinico, G., Hajjat, F. M., & Markos, E. 2017. Information sensitivity typology: Mapping the degree and type of risk consumers perceive in personal data sharing. *Journal of Consumer Affairs*, *51*(1): 133-161.

Mothersbaugh, D. L., Foxx, W. K., Beatty, S. E., & Wang, S. 2012. Disclosure antecedents in an online service context: The role of sensitivity of information. *Journal of Service Research*, *15*(1): 76-98.

Naumovska, I., & Lavie, D. 2021. When an industry peer is accused of financial misconduct: Stigma versus competition effects on non-accused firms. *Administrative Science Quarterly*, *66*(4): 1130-1172.

Naumovska, I., & Zajac, E. J. 2022. How inductive and deductive generalization shape the guilt-by-association phenomenon among firms: Theory and evidence. *Organization Science*, *33*(1): 373-392.

Navis, C., & Glynn, M. A. 2010. How new market categories emerge: Temporal dynamics of legitimacy, identity, and entrepreneurship in satellite radio, 1990–2005. *Administrative Science Quarterly*, *55*(3): 439-471.

Navis, C., & Glynn, M. A. 2011. Legitimate distinctiveness and the entrepreneurial identity: Influence on investor judgments of new venture plausibility. *Academy of Management Review*, *36*(3): 479-499.

Nissenbaum, H. 2004. Privacy as contextual integrity. *Washington Law Review*, *79*(1): 119-157.

Ohm, P. 2015. Sensitive information. *Southern California Law Review*, *88*(5): 1125-1196.

Paruchuri, S., & Misangyi, V. F. 2015. Investor perceptions of financial misconduct: The heterogeneous contamination of bystander firms. *Academy of Management Journal*, *58*(1): 169-194.

Paruchuri, S., Pollock, T. G., & Kumar, N. 2019. On the tip of the brain: Understanding when negative reputational events can have positive reputation spillovers, and for how long. *Strategic Management Journal*, *40*(12): 1965-1983.

Piazza, A., & Jourdan, J. 2018. When the dust settles: The consequences of scandals for organizational competition. *Academy of Management Journal*, *61*(1): 165-190.

Piazza, A., & Perretti, F. 2015. Categorical stigma and firm disengagement: Nuclear power generation in the United States, 1970–2000. *Organization Science*, *26*(3): 724-742.

Pontikes, E. G. 2012. Two sides of the same coin: How ambiguous classification affects multiple audiences' evaluations. *Administrative Science Quarterly*, *57*(1): 81-118.

Pontikes, E. G., & Barnett, W. P. 2017. The non-consensus entrepreneur: Organizational responses to vital events. *Administrative Science Quarterly*, *62*(1): 140-178.

Smith, B. R., Bergman, B. J., & Kreiner, G. E. 2021. When the beacon goes dark: Legitimacy repair work by subsequent actors in an emerging market category. *Journal of Business Venturing*, *36*(5): 106144.

Soublière, J.-F., & Gehman, J. 2020. The legitimacy threshold revisited: How prior successes and failures spill over to other endeavors on Kickstarter. *Academy of Management Journal*, *63*(2): 472-502.

Stinchcombe, A. L. 1965. Social structure and organizations. In J. G. March (Ed.), Handbook of organizations: 142-193. Chicago: Rand McNally.

Suchman, M. C. 1995. Managing legitimacy: Strategic and institutional approaches. *Academy of Management Review*, *20*(3): 571-610.

Vergne, J.-P. 2012. Stigmatized categories and public disapproval of organizations: A mixed-methods study of the global arms industry, 1996–2007. *Academy of Management Journal*, *55*(5): 1027-1052.

Vergne, J.-P., & Wry, T. 2014. Categorizing categorization research: Review, integration, and future directions. *Journal of Management Studies*, *51*(1): 56-94.

Wry, T., Lounsbury, M., & Jennings, P. D. 2014. Hybrid vigor: Securing venture capital by spanning categories in nanotechnology. *Academy of Management Journal*, *57*(5): 1309-1333.

Yu, T., Sengul, M., & Lester, R. H. 2008. Misery loves company: The spread of negative impacts resulting from an organizational crisis. *Academy of Management Review*, *33*(2): 452-472.

Zavyalova, A., Pfarrer, M. D., Reger, R. K., & Shapiro, D. L. 2012. Managing the message: The effects of firm actions and industry spillovers on media coverage following wrongdoing. *Academy of Management Journal*, *55*(5): 1079-1101.

Zheng, Y., & Wang, Q. 2025. Blood diamonds? *Responses of open-source software developers to the Facebook–Cambridge Analytica scandal. Strategic Management Journal*, *46*(7): 1790-1827.

Zimmerman, M. A., & Zeitz, G. J. 2002. Beyond survival: Achieving new venture growth by building legitimacy. *Academy of Management Review*, *27*(3): 414-431.

Zuckerman, E. W. 1999. The categorical imperative: Securities analysts and the illegitimacy discount. *American Journal of Sociology*, *104*(5): 1398-1438.

Appendix

Table A1. Variable Definitions and Measurement

Type	Variable	Definition and measurement	Source
Dependent variable	Funded	1 if total funding recorded in Crunchbase (U.S. dollars) exceeds one dollar, 0 otherwise (firms without a funding record are recorded as 1 in the data)	Crunchbase funding history
Independent variable	Social Platform	1 if industry classifications include at least one of Social Network, Social Media, Private Social Networking, or Dating	Crunchbase industry classification
Moderator	Personal Record (Wellness/Health)	1 if industry classifications include at least one of Wellness, Fitness, Mental Health, Health Care, Personal Health, Health Diagnostics, or Medical	Crunchbase industry classification
Moderator	Content Sharing (Media & Entertainment)	1 if industry classifications include Media and Entertainment	Crunchbase industry classification
Explanatory variable (H4)	User Exposure Control	1 if the company description contains at least one expression in Table A3, or if the firm is classified as Private Social Networking	Crunchbase descriptions (short and full)
Control	ln # Patents	Log of (number of patents + 1)	Crunchbase
Control	Silicon Valley / New England	1 if headquartered in Silicon Valley or New England, respectively	Crunchbase location
Control	Female / White founder	1 if a female or white founder, respectively, is recorded in founder diversity information	Crunchbase Diversity Spotlight
Control	Nonprofit	1 if the firm is a nonprofit organization	Crunchbase
Control	Pledge 1%	1 if the firm participates in Pledge 1%	Crunchbase
Control	# Industry categories	Number of Crunchbase industry classifications registered by the firm	Crunchbase industry classification
Fixed effects	Industry group	43 Crunchbase industry-group dummies (excluding Internet Services, Media and Entertainment, Health Care, and Community and Lifestyle)	Crunchbase industry groups
Fixed effects	Founding year	Founding-year dummies	Crunchbase founding year

Table A2. Robustness Check: Full Sample Including Firms Founded in 2018 (N = 61,618)

Hypothesis	β	*p*
H1 Social platform	−0.256	< 0.001
H2 Social platform × Personal record	−0.692	0.007
H3 Social platform × Content sharing	0.328	0.142
H4 User exposure control (N = 2,104)	0.324	0.104

Note: Logistic regression, including the control variables, industry-group fixed effects, and founding-year fixed effects.

Table A3. Expressions Used to Measure User Exposure Control

Design type	Meaning	Expressions
Anonymous or pseudonymous participation	Users can participate without revealing their real names	anonymous, anonymity, anonymously, pseudonym
Invitation-based or private community	Spaces visible only to designated members	invite-only, invite only, invitation-only, invitation only, by invitation, members-only, members only, private communit-, private network, private social, private group, private space, closed communit-, closed network, closed group, safe space, safe place, safe environment, safe community, safe platform
Visibility settings	Users decide what to show and to whom	privacy control, privacy setting, control who, control what, control over, choose who, choose what
Industry classification	Private service by definition	Private Social Networking classification

Note: User exposure control is a design that lets users decide how much, and to whom, they are visible. The list of expressions was defined by design type before any results were examined, and it was fixed after reading the actual descriptions and removing expressions captured in unrelated contexts. All expressions were searched in lowercased descriptions. Firms by type: anonymous or pseudonymous participation 29, invitation-based or private community 26, visibility settings 11, Private Social Networking classification 63. Six firms fall into two or more types, and 121 firms fall into at least one.

Table A4. Results under Alternative Definitions of Social Platforms

Definition of social platform	H1	H2	H3	H4
Main definition	−0.2481***	−0.6402*	0.5436*	0.4378*
	(0.0596)	(0.2841)	(0.2352)	(0.2156)
Excluding Dating	−0.2619***	−0.6386*	0.5589*	0.4674*
	(0.0605)	(0.2913)	(0.2355)	(0.2188)
Social Network and Social Media only	−0.2754***	−0.6492*	0.5623*	0.4239+
	(0.0612)	(0.2986)	(0.2379)	(0.2537)

Note: Logistic regression, including the control variables, industry-group fixed effects, and founding-year fixed effects. Standard errors in parentheses. H1 is the main effect of social platform status; H2–H3 are interaction terms; H4 is estimated within social platforms under each definition. $^{+}p < 0.10$, $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

Table A5. Comparison with and without Industry-Group Fixed Effects (Linear Probability Model)

Variable	(1)	(2)	(3)	(4)
Social platform	−0.075***	−0.049***	−0.044***	−0.055***
	(0.010)	(0.011)	(0.011)	(0.011)
Social platform × Personal record			−0.127*	
			(0.053)	
Social platform × Content sharing				0.110*
				(0.051)
Constant	0.348***	0.349***	0.348***	0.349***
	(0.016)	(0.016)	(0.016)	(0.016)
Controls	Yes	Yes	Yes	Yes
Industry-group FE	No	Yes	Yes	Yes
Founding-year FE	Yes	Yes	Yes	Yes
Observations	51,738	51,738	51,738	51,738
R-squared	0.077	0.113	0.113	0.113

Note: Linear probability model. Robust standard errors in parentheses. N = 51,738. $^{+}p < 0.10$, $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

Table A6. Description-Based Validation of Service Types

Panel A. Share of firms whose descriptions contain each set of expressions, by social platform type			
Social platform type	N	Self-recording / sensitive information	Shared content
Personal record services (Wellness/Health)	71	28.2%	19.7%
Content sharing services (Media & Ent.)	101	4.0%	67.3%
Other social platforms	1,648	2.6%	31.1%
Panel B. Moderating effects of service types measured from description expressions			
Description-based service type	Social platforms with expressions	Main effect	Social platform × service type
Self-recording / sensitive information	67	0.418^{***}	−0.235
		(0.042)	(0.281)
Shared content	594	0.328^{***}	0.249^{*}
		(0.035)	(0.121)

Note: Panel B is a logistic regression (N = 51,738) including the control variables (including the personal record and content sharing classifications), industry-group fixed effects, and founding-year fixed effects. Standard errors in parentheses. Self-recording / sensitive-information expressions: track your/their/my, journal, diary, mood, symptom, habit, personal health, health record, medical record, self-care, sleep, mental health, therapy, therapist, support group, peer support, coping, grief, addiction, anxiety, depression, LGBT, sexual, illness, disease, patient, cancer, diabet-, pregnan-, fertility. Shared-content expressions: stream, video, music, creator, follower, fans, audience, broadcast, influencer, entertain-, podcast, watch. $^{+}p < 0.10$, $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.